

A Model-based Motion Amplifying Projection for Food Elasticity Manipulation

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ABSTRACT

Perceived taste and flavor of food are greatly affected by visual information, and manipulating them in projection mapping (PM) has been explored. In this paper, we developed systems that use a spring-mass-damper model to project visually amplified motion effects of food and the virtual shadow that moves in synchrony with the pudding's motion. Subjectively, it was perceived that the motion amplification and the virtual shadow effect gave the impression of increased elasticity.

Index Terms: Projection mapping, Food, Motion model

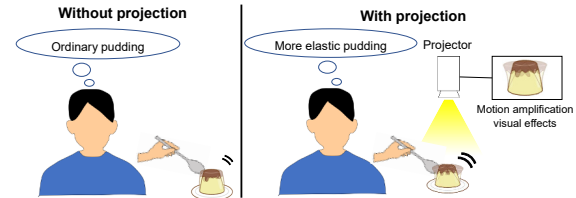
1 INTRODUCTION

All five senses affect the gustatory perception of food, with sight, in particular, known to have a significant impact on the eating experience. Researchers have revealed that changes in the appearance of food modify gustatory perception in projection mapping (PM). Nishizawa et al. revealed that the sweetness of a sponge cake increased with an increase in its chroma, and the flavor of chips was changed by its hue in PM [4]. Fujimoto et al. found that adjusting the saturation and partial color of food enhances the subjective tastiness of the food [1]. Suzuki et al. demonstrated that overlaying bubbling visual effects onto curry increased users' anticipation of the meal [5]. These studies highlight the potential of PM for modifying gustatory perception.

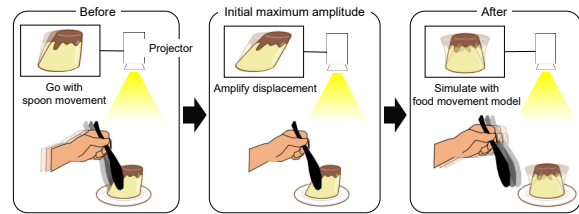
However, it has not been explicitly investigated whether PM-based appearance manipulation of food significantly affects the perceived texture of the food. Specifically, we developed a system that projects visually amplified motion effects to investigate whether PM can influence textural properties such as softness [2]. Our user study results suggested that projecting visually amplified motion effects onto pudding can alter the perception of its softness. However, this method required sprinkling gold powder on the pudding to enable the motion estimation, thus altering the food itself. In this paper, we aim to project visually amplified motion effects without altering the food. Fig. 1(a) illustrates the overview of this study. We use a pudding as the projection target.

2 PROPOSED METHOD

Assuming the user hits the pudding once with a spoon, the system acquires the spoon's coordinates to calculate the pudding's displacement and project visually amplified motion effects. The pudding's motion is divided into two phases: before and after reaching the initial maximum amplitude, defined as the displacement from the initial position to the farthest point the spoon pushes the pudding. An overview of the proposed method is shown in Fig. 1(b). Before reaching the initial maximum amplitude, the system goes



(a) Overview of the motion amplifying projection



(b) Overview of the proposed method

Figure 1: System overview

the projected pattern with the spoon's movement. To achieve a visually amplified motion effect, the initial maximum amplitude is multiplied by a constant factor. After reaching the initial maximum amplitude, the generation of the projection image is based on a spring-mass-damper model approximating the pudding's motion [3].

2.1 Acquisition of Spoon Coordinates

To isolate the food through image processing, we used a black spoon. It is assumed that the spoon moves toward the center of the pudding from its back side, touches the pudding once, and then moves away from the center of the pudding. A mask is first created to extract the pudding region, and binarization is applied within the mask to extract the spoon region. To prevent failure in extracting the spoon region due to the influence of the black plate, the pudding region was eroded and used instead. A bounding box is generated for the spoon region, and the midpoint of the box's edge closest to the center of the pudding is defined as the spoon's coordinates (Fig. 2). These coordinates are tracked from the spoon's contact with the pudding until the pudding reaches its initial maximum amplitude. To detect the moment of contact, the pixel value differences between frames are calculated for a specific region of the pudding, and contact is determined when the difference exceeds a defined threshold. To identify the moment when the pudding reaches its initial maximum amplitude, it is assumed to coincide with the moment the spoon moves away from the center of the pudding. This is determined based on the distance between the pudding's center and the spoon's coordinates.

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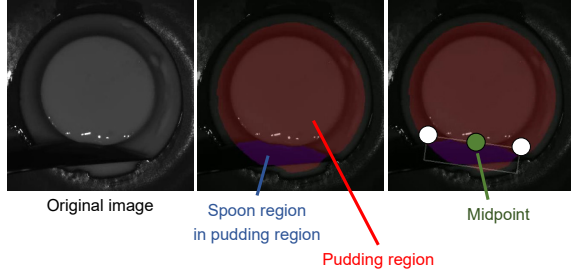


Figure 2: Left: original image, middle: pudding region and spoon region in pudding region, right: spoon region's bounding box and its midpoint

2.2 Pudding Motion Model

The motion of the pudding after it reaches its initial maximum amplitude is approximated using a spring-mass-damper model. Let the pudding's mass be m , its displacement from the equilibrium position be x , the damping coefficient be c , and the spring constant be k . The equation of motion is expressed as follows:

$$m\ddot{x} + c\dot{x} + kx = 0 \quad (1)$$

where it is assumed that the user does not touch the pudding again after making contact with the spoon once, until the pudding's motion ceases. Let A denotes the amplitude of the pudding at the moment it reaches its initial maximum amplitude. The initial conditions for this motion are given as $x(0) = A$ and $\dot{x}(0) = 0$.

The solution to Equation 1, using the initial conditions, is expressed as follows:

$$x(t) = A' e^{-\gamma t} \cos(\omega t - \arctan(\frac{\gamma}{\omega})) \quad (2)$$

$$A' = A \sqrt{1 + (\frac{\gamma}{\omega})^2} \quad (3)$$

where ω and γ are expressed as follows:

$$\gamma = \frac{c}{2m}, \quad \omega = \sqrt{\frac{k}{m} - \gamma^2} \quad (4)$$

Since c and k are unknown, they are determined by obtaining the period and amplitude peaks when the pudding is actually moved. Let the period when the pudding is moved be T , t_1 be the time when the first peak occurs in Equation 2, and X_1 be the amplitude at that time. Let t_{n+1} be the time when the $(n+1)$ -th peak occurs, and X_{n+1} be the amplitude at that time. From Equation 2, γ and ω is expressed as follows:

$$\gamma = -\frac{1}{nT} \log \frac{X_{n+1}}{X_1}, \quad \omega = \frac{2\pi}{T} \quad (5)$$

By substituting Equations 5 into 4, the values of c and k can be determined.

2.3 Creation of the Projected Image with Amplified Motion

By using the displacement of the spoon and the amplitude obtained from the modeled data, the image was translated to create a projected image with amplified motion. The image to be translated was fixed to the one captured before the spoon's intervention. To create a motion-amplified visual effect, the initial maximum amplitude was multiplied by a constant factor for projection.

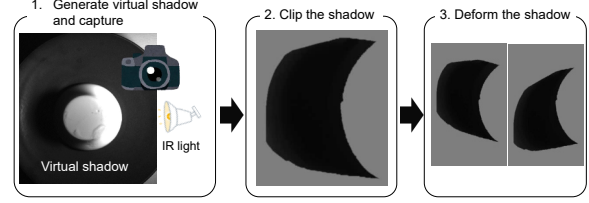


Figure 3: The creation and projection method for the virtual shadow

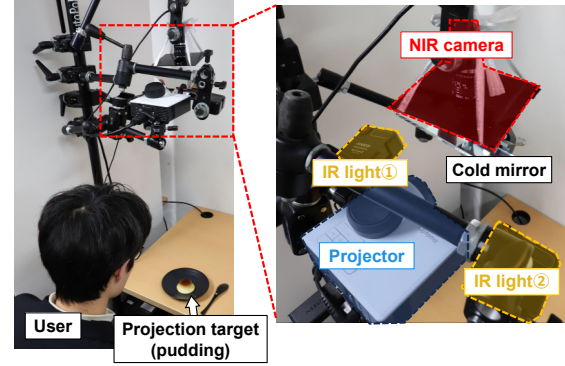


Figure 4: Hardware configuration

2.4 Creation of visual effects for the virtual shadow

We propose a method to enhance the perception of pudding motion by utilizing the virtual shadow. The virtual shadow refers to shadows projected by a projector that do not exist in reality.

The creation and projection method for the virtual shadow is explained in Fig. 3. First, an infrared (IR) light source is used to generate a shadow on the pudding, which is then captured by a camera. Next, a paint application is used to erase all areas except the shadow, extracting only the shadow region to create the projection image manually. To prevent gaps from forming between the pudding and the shadow, a portion overlapping with the pudding is filled in. Using homography transformation, the side closer to the pudding is fixed, while only the side farther from the pudding is moved. By applying the method described in this section, the virtual shadow is moved to synchronize with the pudding, creating the appearance that the shadow is moving in harmony with the pudding. To avoid unnatural appearances caused by the virtual shadow overlapping the pudding itself, the virtual shadow is projected onto areas other than the pudding. As described in section 2.1, the pudding region is extracted using binarization, and this was used to create the projected image of the virtual shadow.

3 IMPLEMENTED SYSTEM

Fig. 4 illustrates the device configuration of the system. The camera (FLIR, FL3-U3-13S2M-CS) captures the movement of the spoon. A PC (CPU: Intel Core i7-12700, RAM: 16 GB, GPU: GeForce RTX 3060) receives the images acquired by the camera, creates a visual effect that amplifies the movement of the food, and then sends the created visual effect to the projector (Optoma ML1050ST+S1J). We use a cold mirror (Edmund Optics, 101×127 mm², 45° AOI) to match the optical axes of the camera and the projector and apply the homography transformation to perform the geometric registration of these devices. A visible light

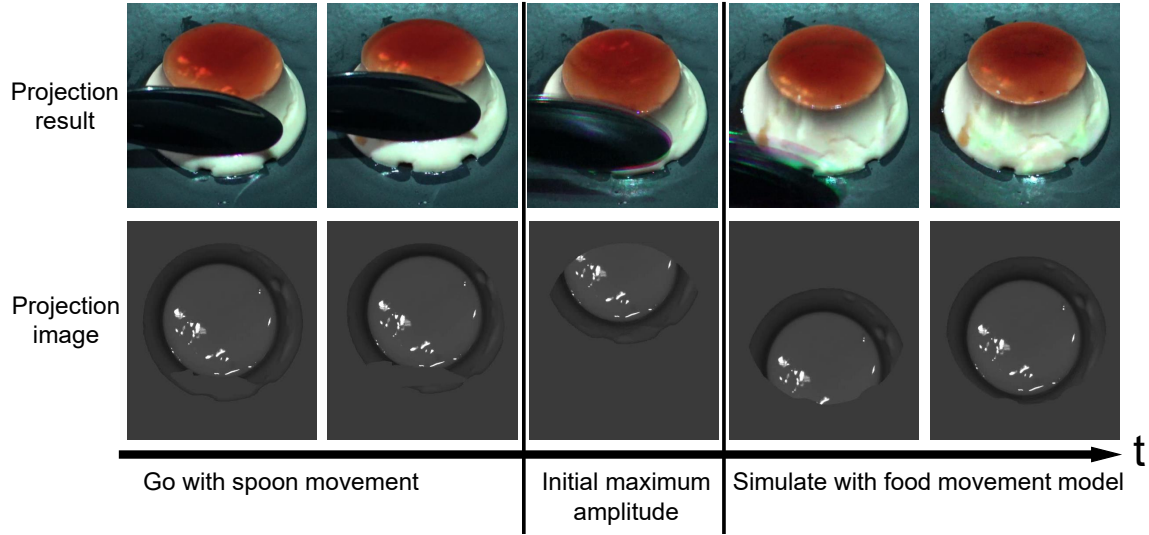


Figure 5: The projection result and image of the motion amplification effect. Top: Projection result; Bottom: Projection image.

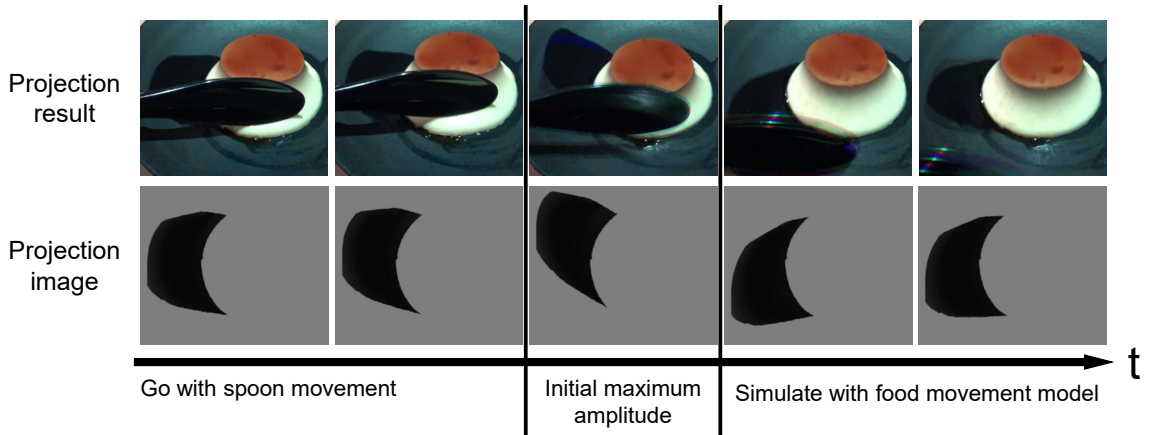


Figure 6: The projection result and image of the virtual shadow effect. Top: Projection result; Bottom: Projection image.

cut/infrared light pass filter is attached to the camera to acquire motion information without being obstructed by the projected light. The food is illuminated by IR light sources.

We used a mass-produced pudding in circulation (Meito Custard Pudding, Kyodo Milk Industry) for the projection target. Using the method described in Section 2.2, c and k for Meito pudding were determined to be $c = 0.87 \text{ kg/s}$ and $k = 137.26 \text{ kg/s}^2$, respectively. The mass m of the pudding was 0.075 kg . The results of projecting images onto the pudding using the motion amplifying method are shown in the top section of Fig. 5, while the projected images are shown in the bottom section of Fig. 5. Here, the initial maximum amplitude was multiplied by a factor of 3 for projection. In the projection images, we fill the non-pudding area with a uniform gray color whose brightness is set to the average brightness of the pudding area. Subjectively, compared to the state without visual effects, the amplified motion appeared to convey a sense of greater elasticity.

The results of projecting images using the virtual shadow projection method are shown in the top section of Fig. 6, while the

projected images are shown in the bottom section of Fig. 6. Here, the initial maximum amplitude was multiplied by a factor of 3 for projection. To enhance the visibility of the virtual shadow, the brightness of the areas outside the virtual shadow is set to 127. The presence of the virtual shadow made the pudding's motion feel more pronounced. However, compared to the direct projection method, the perceived change in the elasticity of the pudding appeared smaller, though this impression remains subjective. This is likely because the motion amplification effect is directly projected onto the pudding itself, creating the impression that the pudding's movement is changing. In contrast, the virtual shadow effect is not directly projected onto the pudding, making the perceived change in motion feel more indirect.

The results of simultaneously projecting the motion amplification method and the virtual shadow are shown in the top section of Fig. 7, while the projected images are shown in the bottom section of Fig. 7. Compared to projecting each effect individually, simultaneous projection appears to give a somewhat stronger impression of increased elasticity. While the virtual shadow alone does not al-

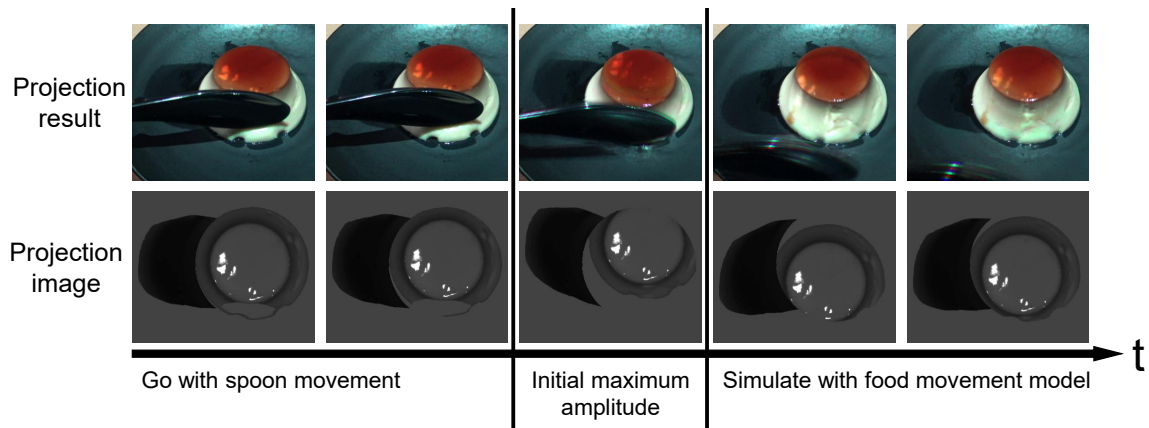


Figure 7: The projection result and image of the motion amplification and the virtual shadow effect. Top: Projection result; Bottom: Projection image.

ter the motion of the pudding itself, resulting in a smaller perceived change in elasticity, projecting both effects together causes the pudding's motion to appear altered and more pronounced.

4 DISCUSSION

There is a delay of approximately 40 ms between the actual motion of the pudding and the motion in the projection. This delay may cause the user to perceive a misalignment or experience discomfort. To address this issue, a method for predicting the spoon's movement to reduce the delay can be considered. Additionally, using a high-speed camera and projector may help minimize the delay.

5 CONCLUSION

In this paper, we proposed a system that utilizes a food movement model to project visually enhanced motion effects of food. Subjectively, the motion amplification and the virtual shadow effect appeared to create a perception of enhanced elasticity. By projecting both effects simultaneously, a greater change in elasticity was perceived. In the future, we plan to conduct user experiments to investigate how the projection of the motion amplification visual effect and the virtual shadow effect influences users' perception of food texture and taste in eating.

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